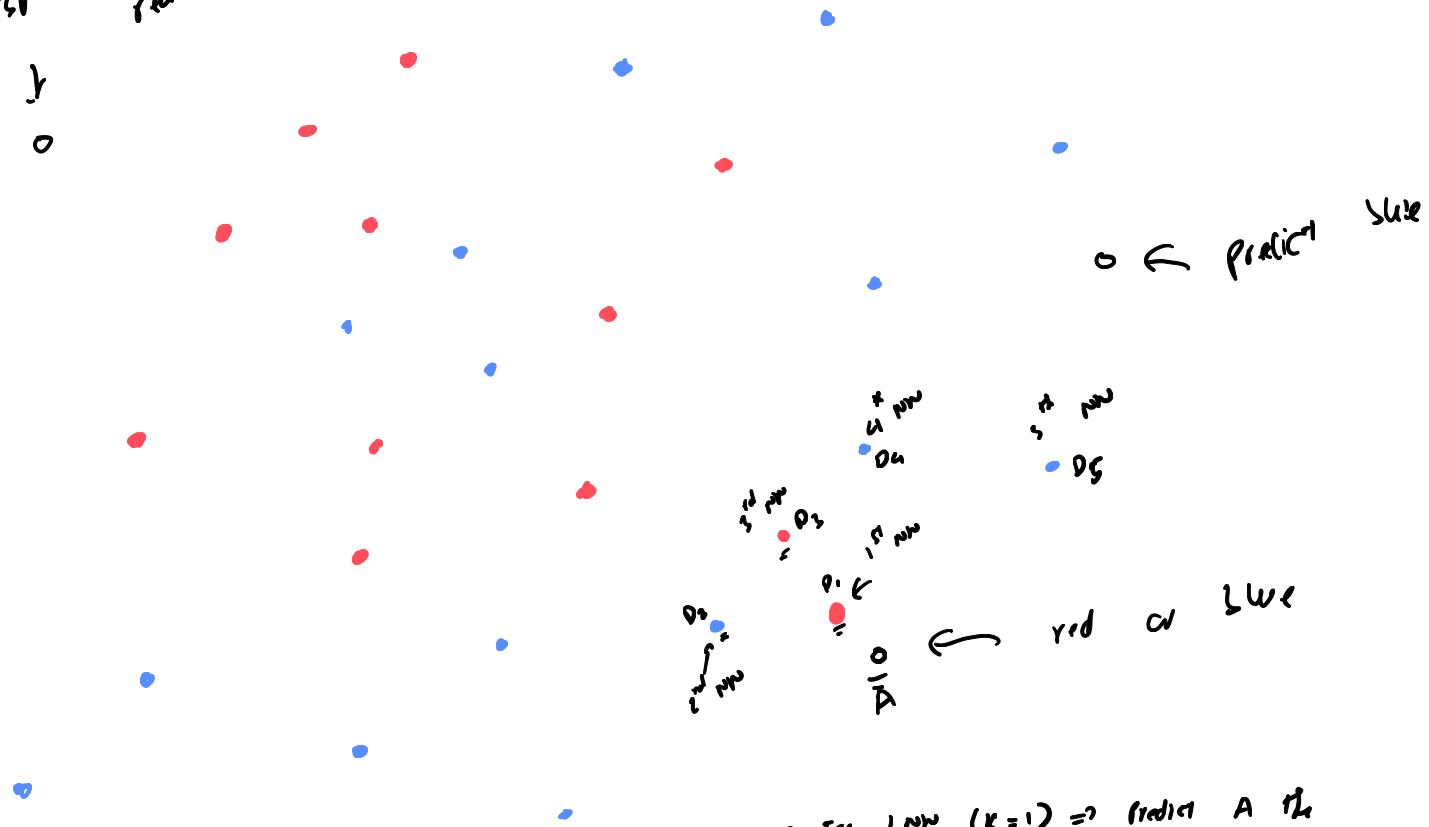
3NN predicts red



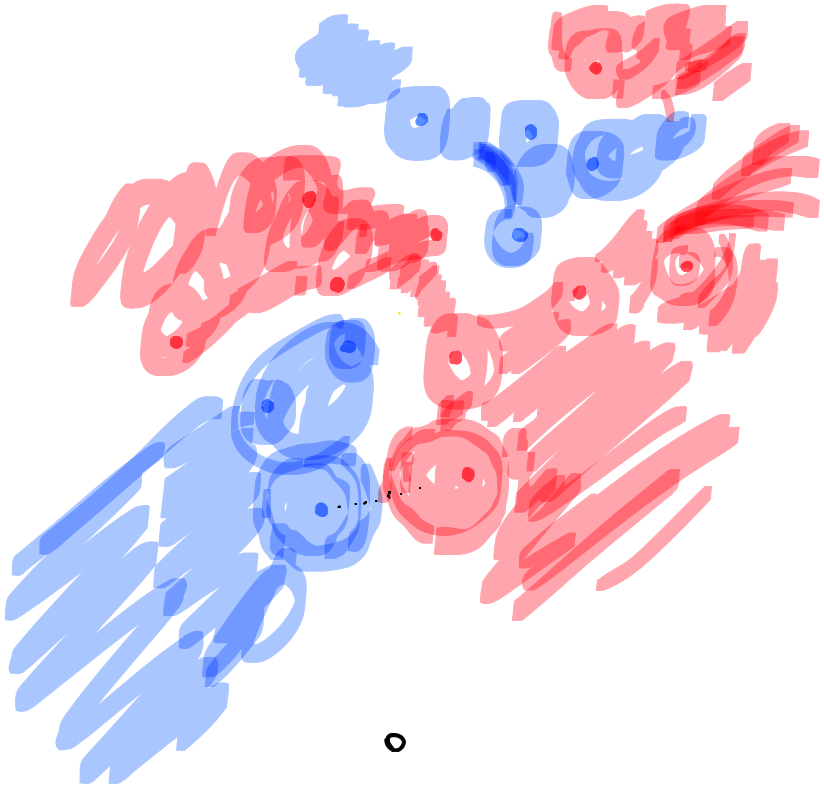
- For 1 NN ($k=1$) ⇒ predict A the same as D_1, red
- For 2 NN ($k=2$) ⇒ predict A : majority of 2 NN 50% red 50% blue
- For 3NN ($k=3$) predict A : majority of (D_1 D_2 D_3) which is red

- Let say there are 27 points (11 reds, 16 blues)

27 NN ($k = 27$)

⇒ As k increases, the model gets weaker.

If k is small, the KNN tends to be "overfit."



9 reds
7 blues

<u>weakest NN</u> 16-NN predict everything is reds

$$\text{Training Accuracy} = \frac{9}{16}$$

<u>strongest NN</u> <u>1 NN</u>

$$\text{Training Accuracy} = \frac{16}{16} = 1$$

<u>10 NN</u>